In [4]:

```
import numpy as np
import pandas as pd
dataset=pd.read_csv("Social_Network.csv")
print(dataset)
dataset=pd.get_dummies(dataset,drop_first=True)
print(dataset)
dataset=dataset.drop("User ID",axis=1)
print(dataset)
dataset["Purchased"].value_counts()
x = dataset[["Age","EstimatedSalary","Gender_Male"]]
y = dataset["Purchased"]
dataset.shape
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1/3,random_state=0)
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
param_grid={
    'criterion':['gini','entropy'],
    'max_features':['sqrt','log2'],
    'max_leaf_nodes':[3,6,12]
}
clf=GridSearchCV(RandomForestClassifier(),param_grid,refit=True,verbose=3,n_jobs=-1)
clf.fit(x_train, y_train)
y_pred=clf.predict(x_test)
print(y_pred)
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test, y_pred)
print(cm)
from sklearn.metrics import classification_report
clf_report = classification_report(y_test, y_pred)
print(clf_report)
```

```
      User ID  Gender  Age  EstimatedSalary  Purchased
0    15624510    Male   19            19000          0
1    15810944    Male   35            20000          0
2    15668575  Female   26            43000          0
3    15603246  Female   27            57000          0
4    15804002    Male   19            76000          0
..        ...     ...  ...              ...        ...
395  15691863  Female   46            41000          1
396  15706071    Male   51            23000          1
397  15654296  Female   50            20000          1
398  15755018    Male   36            33000          0
399  15594041  Female   49            36000          1

[400 rows x 5 columns]
      User ID  Age  EstimatedSalary  Purchased  Gender_Male
0    15624510   19            19000          0         True
1    15810944   35            20000          0         True
2    15668575   26            43000          0        False
3    15603246   27            57000          0        False
4    15804002   19            76000          0         True
..        ...  ...              ...        ...          ...
395  15691863   46            41000          1        False
396  15706071   51            23000          1         True
397  15654296   50            20000          1        False
398  15755018   36            33000          0         True
399  15594041   49            36000          1        False

[400 rows x 5 columns]
     Age  EstimatedSalary  Purchased  Gender_Male
0     19            19000          0         True
1     35            20000          0         True
2     26            43000          0        False
3     27            57000          0        False
4     19            76000          0         True
..   ...              ...        ...          ...
395   46            41000          1        False
396   51            23000          1         True
397   50            20000          1        False
398   36            33000          0         True
399   49            36000          1        False

[400 rows x 4 columns]
Fitting 5 folds for each of 12 candidates, totalling 60 fits
[0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 1 0 1 1 0 0 1 1 1 0 0 1 0 0 1 0 1 0 1 0 0 0 0 1 0 0 1
 0 0 0 0 1 1 1 1 0 0 1 0 0 1 1 0 0 1 0 0 0 1 0 1 1 1 1 1 0 1 1 1 0 0 0 0 0
 0 0 1 1 1 1 0 0 1 0 1 1 1 0 0 1 1 1 1 0 1 0 1]
[[78  7]
 [ 3 46]]
              precision    recall  f1-score   support

           0       0.96      0.92      0.94        85
           1       0.87      0.94      0.90        49

    accuracy                           0.93       134
   macro avg       0.92      0.93      0.92       134
weighted avg       0.93      0.93      0.93       134
```

In [ ]: